

Determinants

Defⁿ Let $V_1, V_2, \dots, V_k, W$ be vector spaces. A function $f: V_1 \times V_2 \times \dots \times V_k \rightarrow W$ is called multilinear if it is linear in each of the k arguments when the remaining arguments are fixed. Eg. in the first slot this means: for each scalar α and for each $v_1, v'_1 \in V_1$, $f(\alpha v_1 + v'_1, v_2, \dots, v_k) = \alpha f(v_1, v_2, \dots, v_k) + f(v'_1, v_2, \dots, v_k)$

Example: the multiplication map $\mathbb{R}' \times \mathbb{R}' \rightarrow \mathbb{R}'$ is bilinear (= multilinear with $k=2$)
 $(a, b) \mapsto ab$

Exercise: The set of all multilinear maps $\text{Mult}(V_1 \times V_2 \times \dots \times V_k, W)$ is a vector space.

We seek a multilinear map "det": $\underbrace{\mathbb{R}^n \times \mathbb{R}^n \times \dots \times \mathbb{R}^n}_{n \text{ copies}} \rightarrow \mathbb{R}$, $A = \underbrace{[v_1, v_2, \dots, v_n]}_{\substack{\text{square matrix } A \\ \text{with columns } v_1, \dots, v_n}} \mapsto \det A$

Theorem: There is exactly one function from $n \times n$ matrices to $\mathbb{R}$ that

(I) is multilinear in columns (II) vanishes when two adjacent columns are equal (III) is 1 on I .
 $\downarrow$
We call such functions "alternating"

Strategy: First prove uniqueness by showing a forced formula. Then show that this formula satisfies I, II and III.

Effect of column operations on any multilinear alternating f (i.e. f satisfying I and II)

(1) Scaling a column scales f by same scalar: $f(\dots, cv_i, \dots) = c f(\dots, v_i, \dots)$ by linearity in slot i

(2) Exchanging any two columns changes sign of f . We do this in two steps

• Swapping adjacent columns: let's show $f(v_1, v_2, v_3, \dots, v_n) = -f(v_2, v_1, v_3, \dots)$

Standard trick: consider $f(v_1 + v_2, v_1 + v_2, v_3, \dots)$. This is 0 by II and also equals sum of four terms by linearity in first two slots. Two of the four terms are 0 by II, giving the result.

[For any adjacent slots $i, i+1$ the same argument works.]

• Swapping any two columns $i < j$. This can be achieved by a succession of adjacent swaps. How many? Answer $2(j-i) - 1$

Reason: move v_i to the right using $j-i$ adjacent swaps till v_i reaches slot j . Then move v_j to the left using $j-i-1$ adjacent swaps to slot i

$$\text{Thus } f(\dots, v_i, \dots, v_j, \dots) = -f(\dots, v_j, \dots, v_i, \dots) \quad \checkmark$$

(3) $v_i \rightarrow v_i + c v_j$ leaves f unchanged

$$f(\dots, v_i + c v_j, \dots, v_j, \dots) = \underbrace{f(\dots, v_i, \dots, v_j, \dots)}_{\text{By linearity in slot } i} + c \underbrace{f(\dots, v_j, \dots, v_j, \dots)}_{0 \text{ by (2)}}$$

Upshot: I get a formula for any such f as follows. Take $n \times n$ matrix $A = (a_{ij})$

$$f(A) = f\left(\sum_{i=1}^n a_{i1} e_i, \sum_{i=1}^n a_{i2} e_i, \dots\right) \text{ where } e_1, e_2, \dots, e_n = \text{std basis of } \mathbb{R}^n$$

$$= \sum_{\substack{1 \leq \sigma_j \leq n \\ \text{for } j=1, \dots, n}} a_{\sigma_1 1} a_{\sigma_2 2} \dots a_{\sigma_n n} f(e_{\sigma_1}, e_{\sigma_2}, \dots, e_{\sigma_n})$$

Sum of n^n terms

Unless $\{\sigma_1, \sigma_2, \dots, \sigma_n\} = \{1, 2, \dots, n\}$ i.e. unless $\sigma_1 \sigma_2 \dots \sigma_n$ is a permutation of $1 2 \dots n$, $f(e_{\sigma_1}, e_{\sigma_2}, \dots, e_{\sigma_n}) = 0$ because some e_i repeats inside f . Therefore

$$f(A) = \sum_{\substack{\sigma = \sigma_1 \sigma_2 \dots \sigma_n \\ \text{permutation of } 1, \dots, n}} a_{\sigma_1 1} a_{\sigma_2 2} \dots a_{\sigma_n n} f(e_{\sigma_1}, e_{\sigma_2}, \dots, e_{\sigma_n})$$

Sum of $n!$ terms

and each $f(e_{\sigma_1}, e_{\sigma_2}, \dots, e_{\sigma_n}) = (-1)^S f(\text{Identity matrix})$
where S = number of adjacent swaps used

to convert $\sigma_1 \sigma_2 \dots \sigma_n$ to $1 2 \dots n$

This proves uniqueness of determinant function
For existence the parity of S above must be independent of the sequence of adjacent swaps used to convert $\sigma_1 \sigma_2 \dots \sigma_n$ to $1 2 \dots n$. We will prove existence separately by using a particular sequence, which will in turn imply the stated independence.

Defⁿ Permutation = bijection from $\{1, \dots, n\}$ to itself

Set of permutations of $\{1, \dots, n\}$ is denoted S_n

Observe: Composition and inverse of permutations are permutations

Notation: permutation $\sigma =$ sequence $\sigma_1 \sigma_2 \dots \sigma_n$
 $=$ function $\begin{cases} 1 & 2 & \dots & n \\ \sigma_1 & \sigma_2 & \dots & \sigma_n \end{cases}$

Defⁿ Length of permutation σ , denoted $l(\sigma)$

$$= \{ (i, j) \mid i < j \text{ and } \sigma_i > \sigma_j \}$$

$=$ number of "inversions" / "bad pairs"

Observe that there is a way to reorder σ to $1 \dots n$ using $l(\sigma)$ swaps. e.g. get 1 in 1st slot, then 2 in 2nd slot etc.

Pf of existence: Define

$$\det(A) = \sum_{\sigma = \sigma_1, \sigma_2, \dots, \sigma_n} a_{\sigma_1 1} a_{\sigma_2 2} \dots a_{\sigma_n n} (-1)^{l(\sigma)}$$

permutation of $1, \dots, n$ Sum of $n!$ terms

This satisfies I, III easily. For II, pair up the $n!$ terms as follows

$$\begin{array}{ccccccc} \sigma_1 & \sigma_2 & \dots & \sigma_i & \sigma_{i+1} & \dots & \sigma_n = \sigma \\ \sigma_1 & \sigma_2 & \dots & \sigma_{i+1} & \sigma_i & \dots & \sigma_n = \sigma' \end{array} \quad \begin{array}{l} \nearrow \\ \searrow \end{array} \text{ pair them}$$

Coeffs in sum corresp to σ, σ' are equal b/c i^{th} col of $A = (i+1)^{\text{th}}$ col of A .

! $l(\sigma) = l(\sigma') \pm 1$ Thus the terms cancel out.

Observe ① $l(\sigma) = l(\sigma')$

② By reindexing we get

$$\det(A) = \sum_{\mu} a_{1\mu_1} a_{2\mu_2} \dots a_{n\mu_n} (-1)^{l(\mu)} \quad (*)$$

③ If we wanted a function with properties I, II, III for rows, then parallel development would give me $(*)$, which we saw $= \det(A)$.

$$\Rightarrow \bullet \det(A) = \det(A^t)$$

• $\det$ is multilinear & alternating in row vectors.

(4) If a row or col is 0, then $\det = 0$

$\therefore$ If cols of A are (dependent $\Rightarrow$ do not span)
then $\det(A) = 0$ b/c $A \xrightarrow{\text{col op}^n} \begin{pmatrix} \text{matrix with} \\ \text{a 0 col} \end{pmatrix}$

and elem row/col opⁿ do not affect whether $\det = 0$.

Cor A is invertible $\Leftrightarrow \det(A) \neq 0$

Pf. A invertible $\Leftrightarrow \text{RREF}(A) = I \Leftrightarrow \det(A) \neq 0$

$\downarrow$
Saw earlier

$\downarrow$
By highlighted observation
(RREF(A) is either I or it has
a zero row.)

Thm $\det(AB) = \det(A) \det(B)$

2 cases (1) $\det(A) = 0$ RHS = 0

cols of $AB = \text{lin combos of cols of } A$

$\therefore \text{span}(\text{cols of } AB) \subset \text{span}(\text{cols of } A)$

$\therefore \det(A) = 0 \Rightarrow \text{cols}(A) \text{ do not span } \mathbb{R}^n$
 $\Rightarrow \text{cols}(AB) \text{ ——— // ———}$

$\Rightarrow \det(AB) = 0 = \text{LHS}$

(2) $\det(A) \neq 0$ then

$f(x) = \frac{\det(Ax)}{\det(A)}$ satisfies I, II, III

$\therefore$ by uniqueness $f(x) = \det(x)$ i.e. $\frac{\det(Ax)}{\det(A)} = \det(x)$

Claim Usual recursive formula for developing
det by any row or any column is valid.

Pf Use our formula and group terms

Formula involving cofactors

Notation for square matrix $A = (a_{ij})$

$A_{ij} = \det(A \text{ with } i^{\text{th}} \text{ row \& } j^{\text{th}} \text{ col erased})$

$\text{adj } A$ is an $n \times n$ matrix $\left[(-1)^{i+j} A_{ji} \right]^{\text{transpose}}$

Exercise $A(\text{adj } A) = \begin{bmatrix} \det A & 0 \\ 0 & \det A \end{bmatrix} = (\text{adj } A) A$ $\rightarrow$

In the next class
We saw the reasoning
that makes this work
by using the 3×3 case

This gives for invertible A

$$A^{-1} = \frac{1}{\det A} (\text{adj } A)$$

e.g. $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ $[A_{ij}] = \begin{bmatrix} d & c \\ b & a \end{bmatrix}$

$$\text{adj } A = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

